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Paul Kelly

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OBJECTIVE

To continue research and increase my experience in the silicon cybersecurity industry

EDUCATION

Rochester Institute of Technology

Bachelor of Science in Computer Engineering

Rochester, NY

Graduation: May 2022

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, ARM and x86 Assembly, VHDL, C#, Java/Smali, Golang

Software: Binary Ninja, Bash, Burp Suite, Windows, Linux, Vim

Technologies: SEM/FIB, FPGA, ESP32, SDR, Bluetooth and BLE, 802.1x, 3G/4G/5G Cellular, Signal Analyzer

EXPERIENCE

IOActive, Inc

Senior Security Consultant

Seattle, WA

June 2022 - Current

- Silicon - Decapping, delayering, and netlist extractions on integrated circuits to identify vulnerabilities or stolen IP
- Automotive - Total vehicle assessments to identify weaknesses in inter-module communications, gateway routing, or general packet handling
- Medical - Security assessments spanning the entire industry of new-age medical equipment including MRI machines, Ultrasound scanners, X-rays, and surgical robots
- Cellular - Utilize SDR to run 3G/4G/5G base stations in the field to assess security posture from rogue base stations.
- Reverse Engineering - Aided by experience with broad range of binary file formats and ISAs, both standard and custom
- Threat Modeling - Performed threat modeling of embedded/automotive systems using STRIDE and attack-surface analysis, identifying trust-boundary violations, exploitable attack paths, and security controls across firmware, diagnostic interfaces, and network protocols.
- Exploit Development - Python and C programs to take advantage of stack overflows, format string bugs, object deserialization, and more

Saab Defense and Security

Electrical Engineer Intern

Syracuse, NY

May 2019 - Dec 2019

- Ran validation processes for RF hardware including writing VHDL test benches, algorithm confirmation MATLAB scripts, and instrumented test equipment to generate mixer intermodulation tables and analyze harmonics.

PROJECTS

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| Silicon ROM Extraction | • Decapped, delayered (through polishing and ion beam milling), and imaged region of interest to rebuild bit mask ROM into byte code that can be reverse engineered through traditional means |
| Silicon Counterfeits | • Decapped, delayered, and imaged silicon chips to extract netlists and identify stolen logic blocks from copyrighted designs |
| MIPS Processor Model | • Designed the instruction fetch, instruction decode, execute, memory access, and write back stages of the MIPS processor in VHDL |
| Links | • Github: https://github.com/pkelly916/
• Website: http://pauljkelly.net |
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